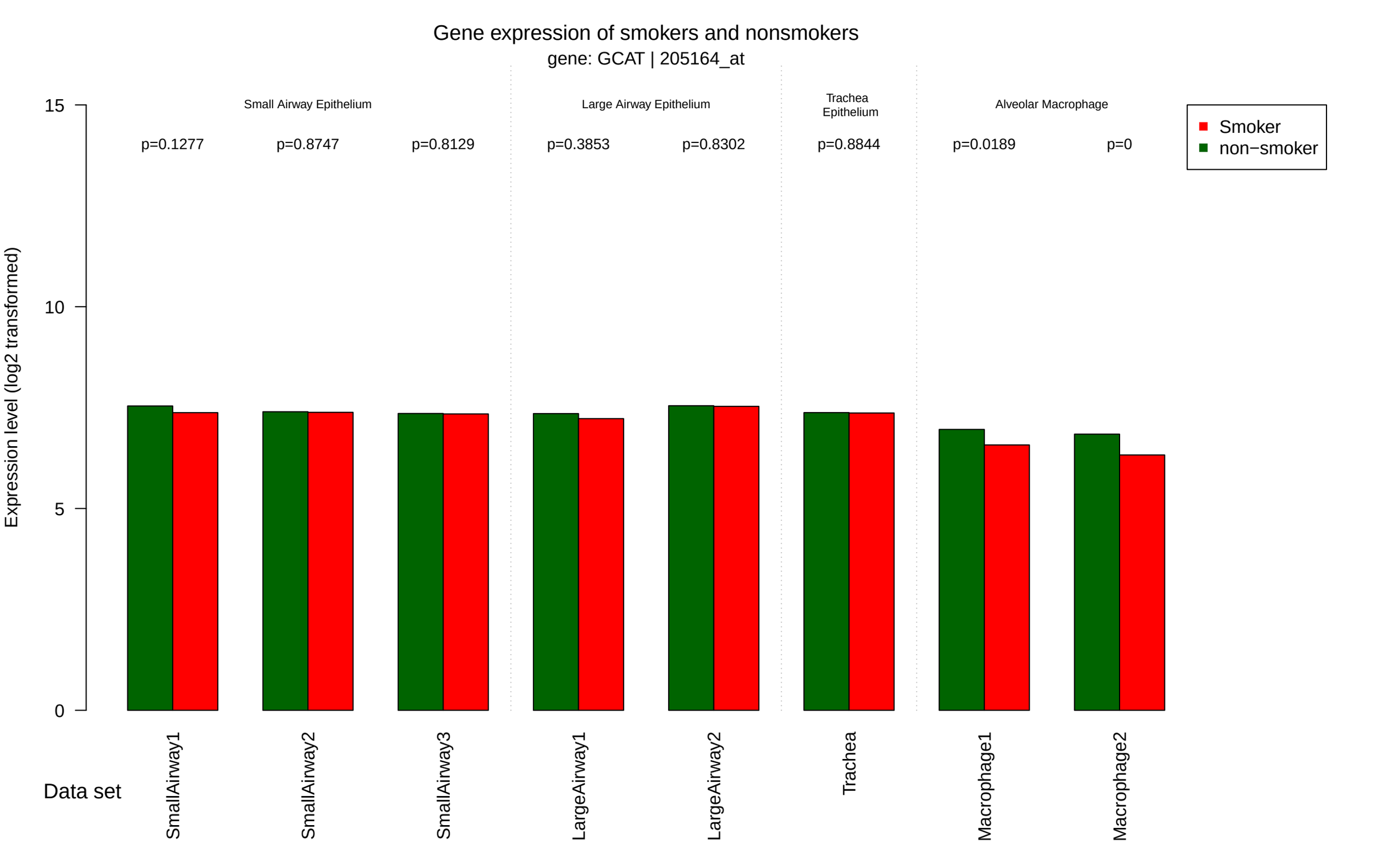
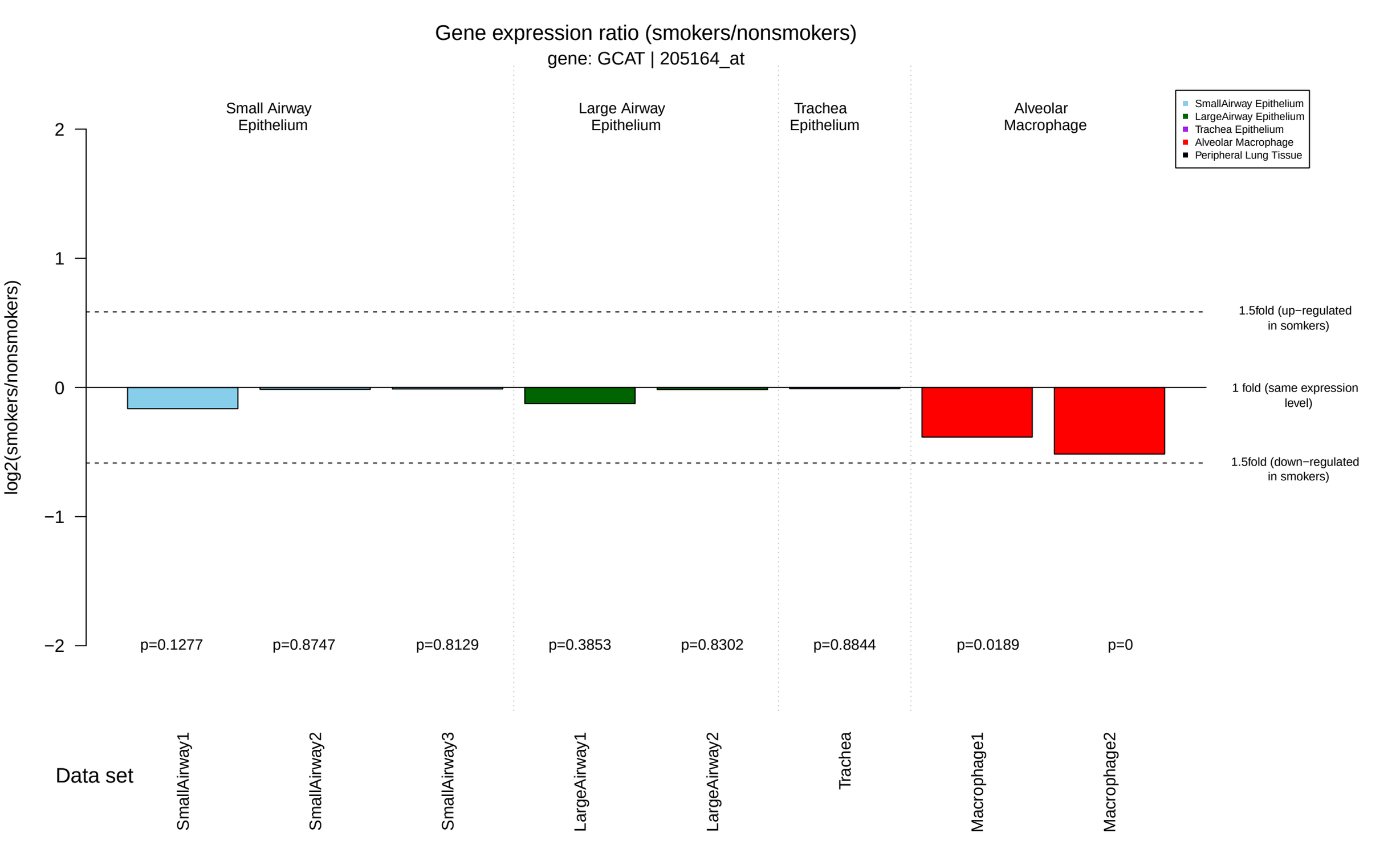
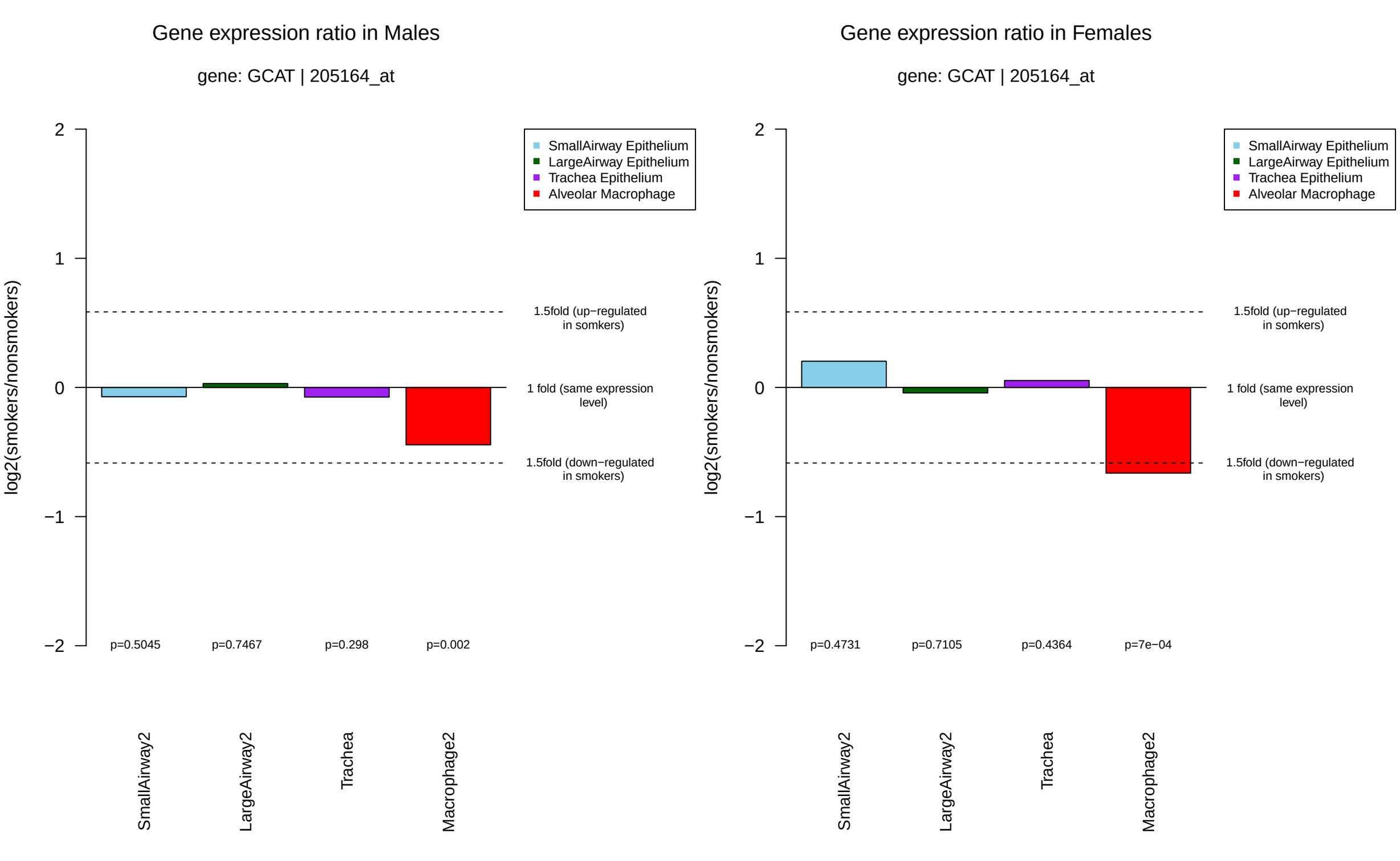
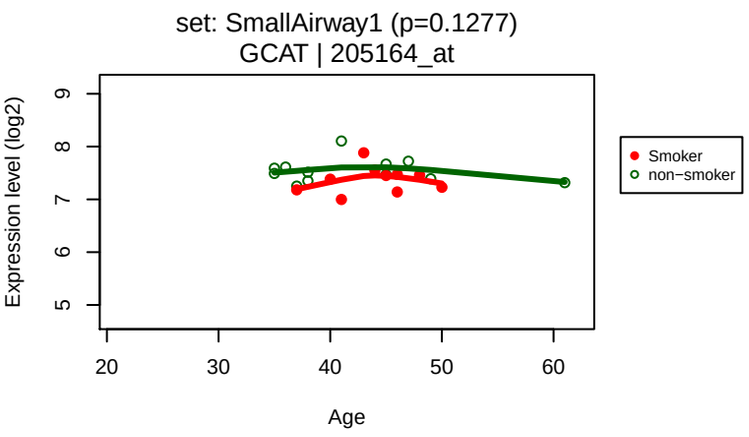
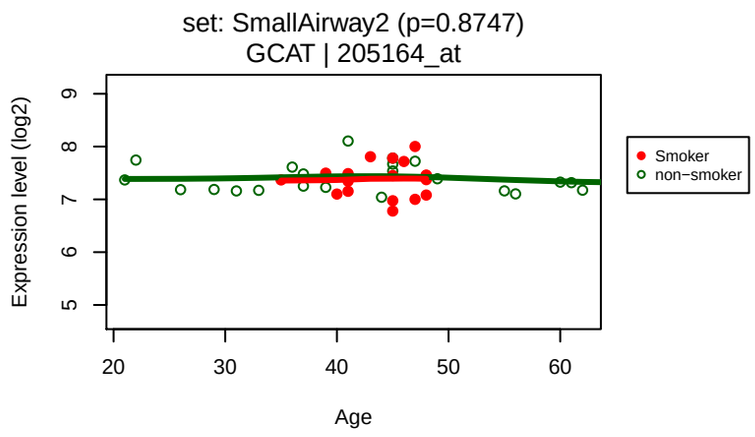
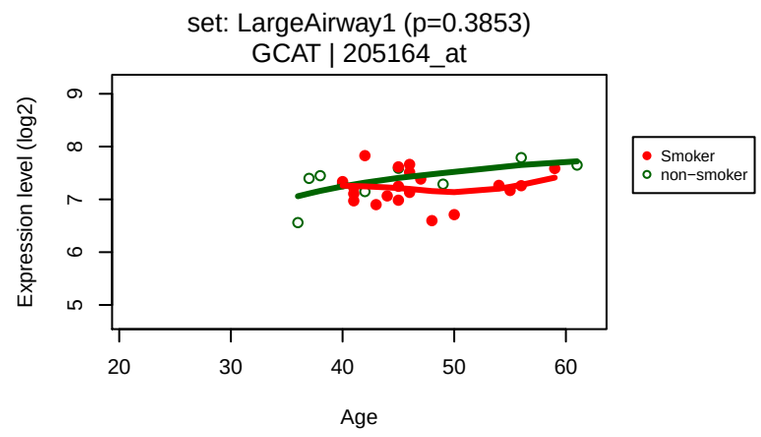

Gene expression of smokers and nonsmokers
gene: GCAT | 205164_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1277
p=0.8747
p=0.8129
p=0.3853
p=0.8302
p=0.8844
p=0.0189
p=0
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

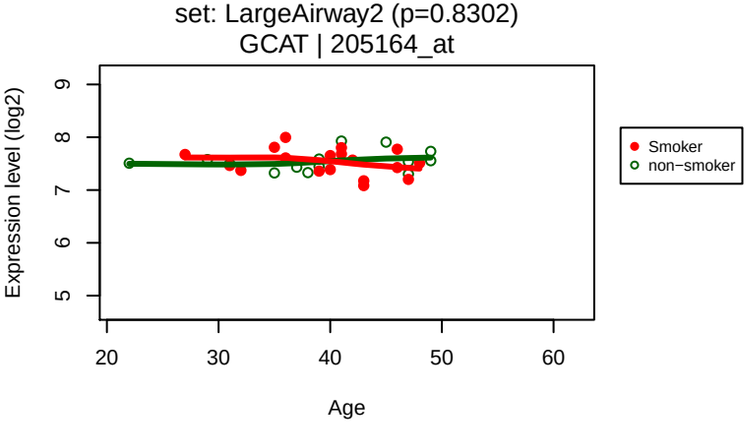

Gene expression ratio (smokers/nonsmokers)
gene: GCAT | 205164_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1277
p=0.8747
p=0.8129
p=0.3853
p=0.8302
p=0.8844
p=0.0189
p=0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

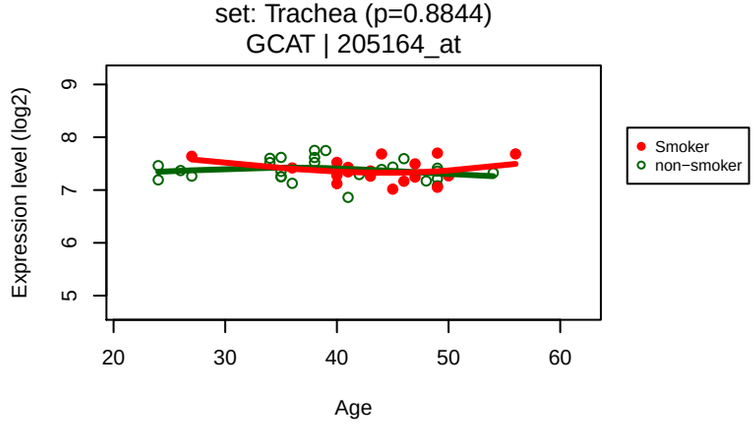

Gene expression ratio in Males
gene: GCAT | 205164_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5045
p=0.7467
p=0.298
p=0.002
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCAT | 205164_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4731
p=0.7105
p=0.4364
p=7e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2

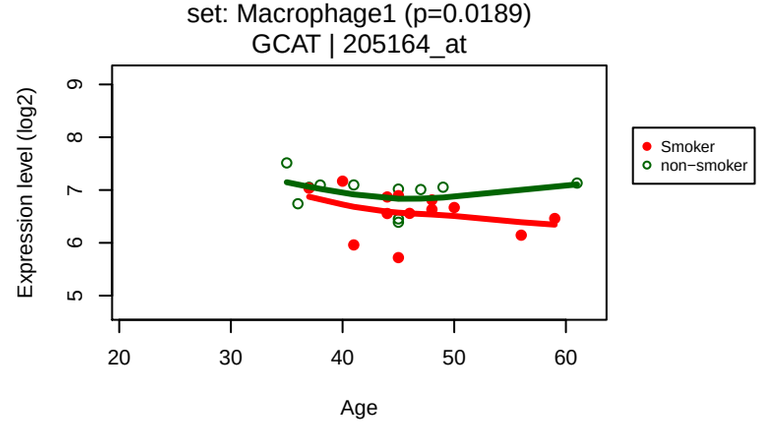

set: SmallAirway1 (p=0.1277)
GCAT | 205164_at
Expression level (log2)
Age
Smoker
non−smoker

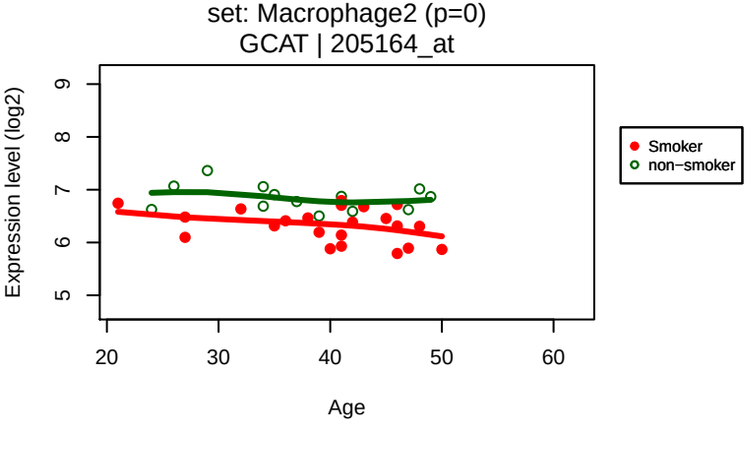

set: SmallAirway2 (p=0.8747)
GCAT | 205164_at
Expression level (log2)
Age
Smoker
non−smoker

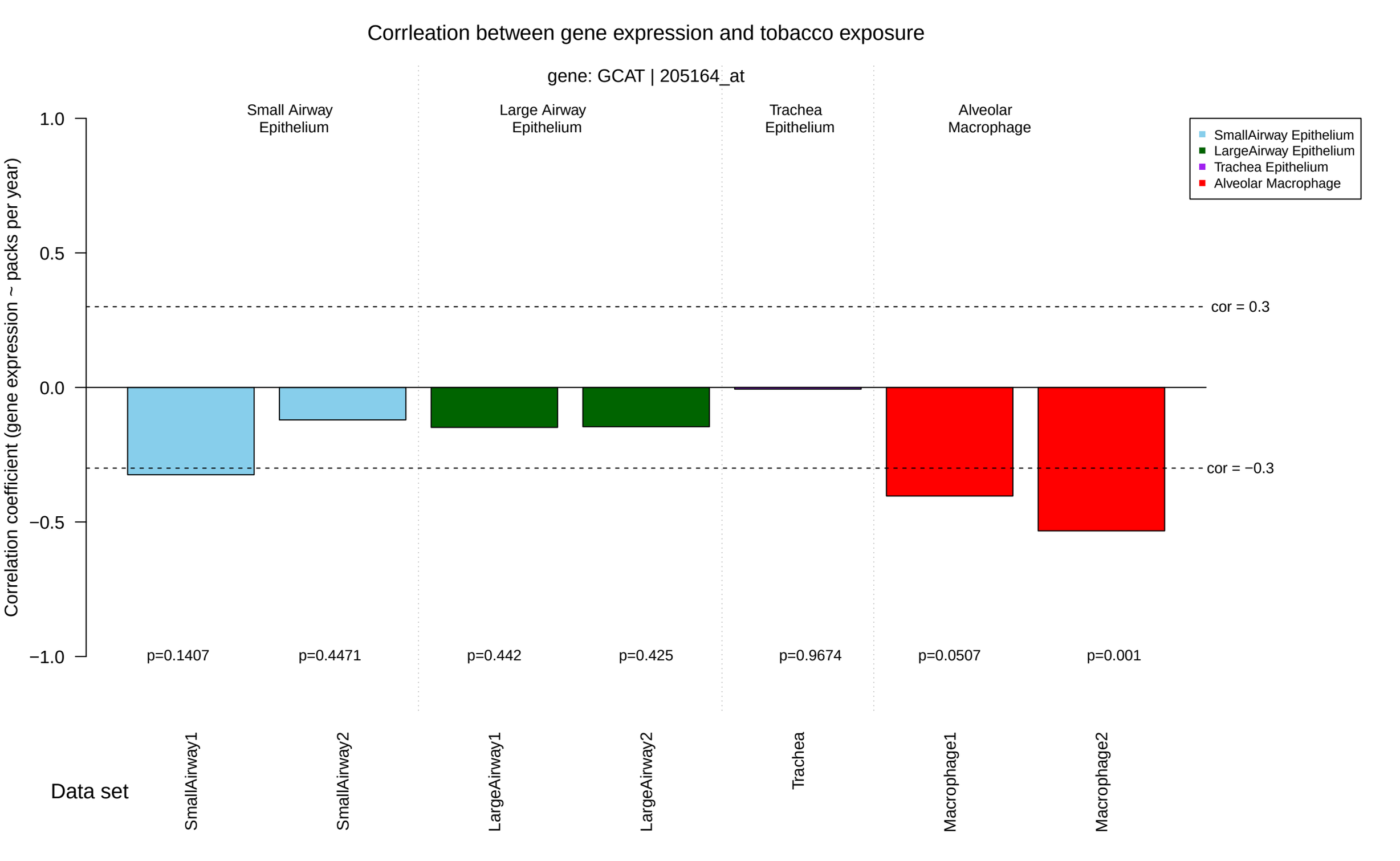

set: LargeAirway1 (p=0.3853)
GCAT | 205164_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8302)
GCAT | 205164_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8844)
GCAT | 205164_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0189)
GCAT | 205164_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0)
GCAT | 205164_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCAT | 205164_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1407
p=0.4471
p=0.442
p=0.425
p=0.9674
p=0.0507
p=0.001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2